

Lesson 4-1 · Period 3

Reciprocal Function + Translations —
Assessment-Critical

Klimsara-adapted · Student-centered · No Blooket

Do Now — Bridge from P2

Algebra 2 · Lesson 4-1 · Period 3

DOK 1

5 min

bridge from P2

Practice #13 · Generalize k For an inverse variation, write an equation that gives the value of k in terms of x and y .

Then predict: Sketch what you think $y = 1/x$ looks like.
(No wrong answers — just predict. We check in Launch.)

Launch — Ex 4 + Ex 5 (teacher models both)

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DOK 2

12 min

teacher models

Parent: $f(x) = 1/x$

- ▶ V.A.: $x = 0$ H.A.: $y = 0$
- ▶ Domain: $x \neq 0$ Range: $y \neq 0$
- ▶ Hyperbola, Quadrants I and III. Never touches either axis.

Translated: $g(x) = 1/(x - h) + k$

- ▶ V.A. $\rightarrow x = h$; H.A. $\rightarrow y = k$
- ▶ Domain: $x \neq h$ Range: $y \neq k$

Explore — TPS: Parent Forms

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DOK 2

35 min

student work

6 items total. Parent forms first (laps 1–3), then translations (laps 4–6).

Try It 4 · Scaled parent Graph $g(x) = 10/x$. State domain, range, and asymptotes.

Practice #11 · Error Analysis: $y = 5/x$ Describe and correct the student's graphing error.

Practice #18 · Reflected parent: $y = -2/x$ Graph

Explore — TPS: Translations

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DOK 2

35 min

student work (cont.)

Before each translation: identify (h, k) , state asymptotes, then graph.

Try It 5 · Translated: $g(x) = 1/(x + 2) - 4$ Graph
 $g(x) = 1/(x + 2) - 4$. Equations of asymptotes? Domain and range?

Practice #19 · ★ ASSESSMENT CRITICAL
 $g(x) = 1/(x - 2) + 6$. Asymptotes, domain, range, and intercepts.
(~8 min)

Share / Summary — Concept Wrap + Assessment Preview

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DOK 2

5 min

DOK-2 reveal + LEHS preview

Sentence frame · Practice #19 pair share “This graph is the parent $f(x) = 1/x$ translated ____ units ____ and ____ units _____. The asymptotes are $x = \underline{\hspace{1cm}}$ and $y = \underline{\hspace{1cm}}$. The intercepts are _____.”

★ Assessment Signal — Practice #19 is the KEYSTONE REHEARSAL Practice #19 is the keystone rehearsal item for Topic 4 LEHS:

Exit Ticket — Pre-Assessment Confidence Check

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DOK 1-2

3 min

not graded

Complete on your exit ticket

1. I can identify asymptotes of $y = 1/(x - h) + k$.
Confidence: 1 / 2 / 3 / 4 / 5
2. I can describe a translation from $f(x) = 1/x$ to $g(x)$.
Confidence: 1 / 2 / 3 / 4 / 5
3. One thing I want to review before Topic 4 assessment:
_____.